

Yuxuan Ni

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EDUCATION

The University of Alabama Ph.D., Electrical and Computer Engineering Tuscaloosa, AL	Sep 2025 - Present
Carnegie Mellon University M.S. in Engineering & Technology Innovation Management Pittsburgh, PA	Jan 2023 - Dec 2023
UESTC / University of Glasgow B.E. in Communication Engineering; B.E. in Electronics and Electrical Engineering with Communications	Sep 2018 - Jun 2022

SELECTED RESEARCH PROJECTS

Bunker Multi-Sensor SLAM, Localization, and Mapping Research project Prof. Shunqiao Sun and Prof. Igal Bilik The University of Alabama / Ben-Gurion University - Built and maintained a ROS2 UGV pipeline for synchronized camera, LiDAR, 4D radar, INS/RTK, wheel-feedback, and driver-control data in bunker, low-light, and GNSS-degraded runs. - Filtered large LiDAR scans for map construction and real-time localization; reduced a 459.5M-point raw scan stream to a 2.13M-point map and cleaned a 24.29M-point map to 13.56M points. - Tested radar/IMU fusion and sensor-specific localization settings against INS/RTK references for trajectory evaluation. - Compared radar, visual, and LiDAR-inertial localization in a low-light, building-occluded sequence; radar odometry reached 7.16 m ATE RMSE in that run.	Aug 2025 - Present
Learning-Based 4D Radar Odometry and Camera Fusion Research project Prof. Shunqiao Sun The University of Alabama - Extended a learning-based radar odometry codebase for sparse 4D radar point clouds, focusing on controlled module-level experiments. - Tested range/intensity filtering, point multiplication, and camera-pose fusion while keeping the pose backbone fixed for comparison. - Added a small pseudo-correspondence head with 84.1K trainable parameters, about 0.53% of the full model. - In frozen-backbone ablations, reduced correspondence-related losses by about 36% and offset error from 0.409 m to 0.363 m; guarded refinement reduced translation RMSE from 0.752 m to 0.311 m.	Jan 2026 - Present
NXP Radar DDM Decoding Collaborative radar signal-processing project - Implemented a MATLAB SFW/DDM simulation and decoding workflow for 12-TX/12-RX, 1024-sample, 512-chirp radar data. - Added multi-TX de-spreading, range-Doppler fusion, CA-CFAR, and DBSCAN steps for staged target extraction tests. - Tested velocity search and quadratic phase-correction desmearing on close-target cases with 2 m range separation and 40 m/s velocity difference.	Aug 2025 - Present
PRIOR RESEARCH EXPERIENCE	
MVMD-Based CSP for EEG Signal Classification Research assistant Prof. Aimin Jiang Hohai University - Applied MVMD to 1,000+ EEG datasets to study inter- and intra-subject signal variability. - Built CSP/ADMM filtering experiments and compared MLP, LDA, and SVM classifiers for EEG classification. - Improved classification accuracy by about 10% and reduced computational complexity by about 20% in the tested setup.	Jun 2024 - Feb 2025
NGSO Satellite In-line Interference Analysis Research assistant Prof. Jon Peha and Prof. Marvin Sirbu Carnegie Mellon University - Simulated satellite-ground links with TLE data in MATLAB and Python, processing million-scale datasets. - Built simulation variants for different constellation and communication-model assumptions, reducing computational cost by about 60%. - Tested an interference-avoidance algorithm that reduced estimated interference probability by about 30%.	Aug 2023 - Jan 2024
Low-Noise Balanced Photodetector for Single Photon Detection Research assistant Prof. Yunxiang Wang UESTC - Worked on a low-noise balanced photodetector circuit using JFETs and photodiode front-end design. - Selected a differential amplifier structure for a 300 MHz theoretical bandwidth target. - Simulated eight LTspice circuit designs and revised PCB layouts in Altium Designer to reduce oscillation.	Sep 2021 - May 2022
5G Clustering Algorithm Simulation for Ultra-Dense Networks Research project Prof. Yuwei Wang Chinese Academy of Sciences - Implemented a clustering algorithm for 5G ultra-dense networks using path-loss and frequency-management assumptions. - Validated the method in NS3 simulations, with a reported 30% improvement in data reception rate.	May 2021 - Oct 2021
Design and Construction of a Multi-Function Rover Team project Communication group lead - Led the rover communication group and built a low-latency wireless link between the rover and control center. - Implemented route tracking and color recognition functions with custom sensing and control logic. - Coordinated embedded programming, circuit design, and communication-module integration under project schedule constraints.	Feb 2021 - Jun 2021

WORK EXPERIENCE

CMU Center for Technology Transfer and Enterprise Creation Enterprise creation intern Pittsburgh, PA - Prepared customer profiles and interview notes for CMU Bio-robotic Lab market discovery, including 60+ lead interviews and 26 key prospects. - Analyzed market and customer data with RStudio and Excel to connect technical use cases with commercialization evidence. - Supported Phlux product-positioning work through customer interviews and technology use-case summaries.	May 2023 - Aug 2023
Signify (China) Investment Co., Ltd. Hardware engineer intern Shanghai, China - Optimized power-circuit designs with active EMI filters and high-performance capacitors, reducing power consumption by 30%. - Debugged and tested IoT lamp functionality, including lighting accuracy and response-logic issues.	Sep 2022 - Dec 2022
China Mobile Communication Group Co., Ltd. Engineering maintenance intern Changzhou, China - Supported planning and maintenance work for 10+ 5G base stations across three service areas. - Documented field troubleshooting for signal interference, transmission delay, and energy-efficiency issues.	Jul 2020 - Aug 2020

PUBLICATION

Y. Ni and L. Shi, "Specific clustering scenarios and simple algorithm simulation in 5G environment," Proc. SPIE 12087, EIECT 2021, 120870M, 2021.

TECHNICAL SKILLS

Python, MATLAB, C/C++, ROS/ROS2, Linux, PCL, OpenCV, PyTorch, NumPy, STK, NS3, LTspice, Altium Designer.